# Brief A — Technical

PdM Detection Approach for NBI-P03 — for the Maintenance Engineering Team

## Approach

We're not modeling vibration or pressure directly — the available signal is shift-level throughput, temperature, and voltage. We derive three rolling health features per machine, computed independently per machine_id so no cross-contamination between pumps' histories occurs:

**Rolling Mean** (5-shift window) — smooths shift noise to expose the underlying trend.

**Rolling RMS** — root-mean-square of the same window; standard vibration-analysis-style metric, more sensitive to large excursions than a plain mean.

**Rolling Range (max − min)** — a volatility proxy; spikes when a machine's operating regime shifts, often before the new average stabilizes.

Target label: incident_type != "none" (Failing) vs. no incident logged (Healthy). Model: Logistic Regression — chosen over a black-box classifier because outputs need to be defensible when they justify pulling a machine offline; coefficients are directly inspectable.

## Performance

| Metric | Value | Sensor logic implication |
|---|---|---|
| Accuracy | 98.4% | Overall correct-call rate across 730 held-out test shifts |
| Sensitivity (Recall, Failing class) | 95.2% | Of every genuinely failing shift, 95% triggers a flag |
| Precision (Failing class) | 96.6% | Of every flag raised, 96.6% is a real problem |
| False Positives | 5 / 583 Healthy shifts | Unnecessary inspection rate on healthy machines is low |
| False Negatives | 7 / 147 Failing shifts | Residual miss rate — the gap a wider window or added sensor channel would target next |

## Validation Before Deployment

We statistically confirmed the underlying effect before trusting the model on it: a Welch's two-sample t-test comparing NBI-P03's throughput against the rest of the fleet returns $t \approx -73.5$, $p \approx 0$ — the 95% CI for NBI-P03's mean (624–642 barrels) doesn't come close to overlapping the fleet's (982–989 barrels). Weather and grid voltage were checked as confounders via correlation ($r = 0.25$ and $r = 0.04$ respectively against throughput) and ruled out — the model isn't picking up a seasonal or electrical artifact.

**Recommended next step:** deploy the rolling-feature calculation live per machine and alert on predicted-Failing probability > 0.5, reviewed weekly against actual maintenance logs to catch drift in the false-negative rate.

# Brief B — Executive

The Value of Early Warning on Pump NBI-P03 — for the CFO

## The Situation

One pump, at one depot, has been running under-maintained for eleven months. It cost us an estimated 194,000 barrels of throughput this year — effectively the entire operation's lost output for the year came from this single piece of equipment. We built a simple early-warning system, using data we already collect, that would have caught this in its first week instead of its eleventh month.

## What It Actually Does

Think of it as a smoke detector for a pump, not a fire extinguisher. It watches each pump's own recent output pattern and raises a flag the moment that pattern starts looking like the early stage of past breakdowns — before the breakdown itself shows up in a monthly report. Out of 730 shifts we tested it against, it correctly caught 95 out of every 100 real problems, and only raised a false alarm about 1 time in 100 — rare enough that it won't create alert fatigue for the maintenance team.

## The Value

| | |
|---|---|
| **Estimated cost avoided** | ~194,000 barrels/year recovered if this pattern is caught and corrected as early as this system would have caught it — instead of running eleven months before anyone noticed. |
| **Risk reduction** | A single piece of equipment was silently responsible for effectively all of this year's throughput loss. An early-warning check turns a year-long blind spot into a same-week catch, without requiring any new hardware or sensors. |
| **Faster decisions** | Maintenance and operations no longer wait for a quarterly or annual review to discover a problem like this — the flag appears while it's still a small, cheap fix. |

**Bottom line:** this isn't a new system to buy or a new team to hire — it's a way of looking at data we already have, applied consistently instead of only during a year-end review. The cost of building it was a few days of analysis. The cost of not having it was a year of one pump quietly underperforming.

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## Reflection: Switching Between Audiences

The hardest part wasn't simplifying the numbers — both briefs use the same underlying model and the same test set. It was deciding which numbers earn a place in each version. The Engineer needed precision and recall as named, comparable metrics because they change what to tune next. The CFO needed exactly one number that matters — barrels, not percentages — because a 95% recall rate means nothing without translating it into what it prevented. The genuine trap was reflex: my first CFO draft still said "recall" once, out of habit, and had to be caught and rewritten as "caught 95 out of 100 real problems." Writing the Engineer brief first, then rewriting rather than summarizing for the CFO, made the switch easier than trying to write one brief and water it down twice.